

5 (**long**) Let

$$A = \begin{bmatrix} 1 & 0 & a \\ 0 & 2 & 0 \\ a & 0 & 3 \end{bmatrix}.$$

- (a) Find the eigenvalues of A . [10]
- (b) Find a normalized (unit length) eigenvector for the median eigenvalue of A . [10]
- (c) Find the eigenvalues of A^{-1} . [5]
- (d) Find $\det(A^3)$. [5]